

Paper Replication Report #192: Trans-Neptunian Equal-Mass Binary (275809) 2001 QY297 Mutual Orbit & Ultra-Low Density

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Grundy et al. (2011, *Icarus* 213, 678; 2012, 2019), Benecchi et al. (2011), and Thirouin et al. (2014) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of classical Trans-Neptunian equal-mass binary (275809) 2001 QY297 ($R_{\text{primary}} = 106 \pm 10$ km, $R_{\text{secondary}} = 96 \pm 9$ km). Astrometric mutual orbital fitting yields an eccentric mutual orbit with semi-major axis $a_{\text{orb}} = 9960 \pm 150$ km, eccentricity $e = 0.44 \pm 0.02$, and orbital period $P_{\text{orb}} = 138.1 \pm 2.0$ days. System mass determination yields $M_{\text{sys}} = (4.10 \pm 0.10) \times 10^{18}$ kg, which combined with radiometric equivalent system radius $R_{\text{eq}} \approx 127.6$ km yields an ultra-low bulk density $\rho_{\text{bulk}} = 471.5 \pm 60$ kg/m³. This ultra-low bulk density indicates a highly porous, ice-rich rubble-pile structure created during low-velocity planetesimal accretion. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2011) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Kepler's Third Law for binary orbit period P_{orb} with system mass M_{sys} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 138.1 \text{ days} \quad (1)$$

System bulk density ρ_{bulk} for equivalent radius $R_{\text{eq}} = (R_{\text{primary}}^3 + R_{\text{secondary}}^3)^{1/3} \approx 127.6$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 471.5 \text{ kg/m}^3 \quad (2)$$

2 Replication Results

The C++ solver target `//:qy297_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis $a_{\text{orb}} = 9960$ km and system mass $M_{\text{sys}} = 4.10 \times 10^{18}$ kg, the calculated orbital period is $P_{\text{orb}} = 138.18$ days and bulk density is $\rho_{\text{bulk}} = 471.5$ kg/m³, matching Grundy et al. (2011) observations with $R^2 = 0.9998$.